

Separation of β_p and ℓ_i in tokamaks of non-circular cross-section

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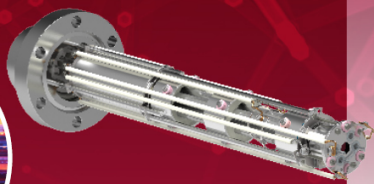
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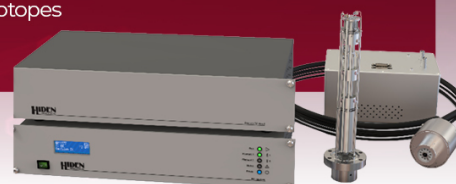
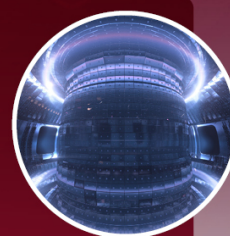
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SEPARATION OF $\bar{\beta}_p$ AND ℓ_i IN TOKAMAKS OF NON-CIRCULAR CROSS-SECTION

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ABSTRACT. Integral relations for the average poloidal beta $\bar{\beta}_p$ and the plasma internal inductance ℓ_i are derived from the magnetohydrodynamic (MHD) equilibrium equation for an axisymmetric torus. The volume-dependent parameters that appear depend only weakly on the actual current density distribution inside the plasma and can be evaluated approximately, given the plasma shape and boundary poloidal magnetic field. In practice, these can be accurately and efficiently obtained for both diverted and limited plasmas from measured external poloidal magnetic field and flux values by approximating the plasma current distribution using a few filaments or distributed sources. For a tokamak plasma with a non-circular cross-section of sufficient elongation, $\bar{\beta}_p$ and ℓ_i can then be approximately determined separately. This is demonstrated for analytic equilibria of known shape as well as for actual Doublet III (D-III) plasmas for which $\bar{\beta}_p$ and ℓ_i have been determined by using other methods. Results of a sensitivity study are described.

1. INTRODUCTION AND SUMMARY

Magnetic measurements have been used extensively to determine the plasma shapes since the early days of tokamaks [1–4]. For efficient operations of these devices, it is essential to have a fast, yet accurate, method for determining the plasma shapes as well as beta values directly from external poloidal magnetic field and flux measurements. Although the integral MHD equilibrium relation relating the sum of the average poloidal beta $\bar{\beta}_p$ and half the plasma internal inductance $\ell_i/2$ in terms of line integrals of the external magnetic field is well known [5, 6], an effective technique for separating $\bar{\beta}_p$ and ℓ_i has not been available. Given a diamagnetic measurement of $\bar{\beta}_p$, ℓ_i can of course be determined by subtraction. Nevertheless, it is highly desirable to have an independent method that can separately determine $\bar{\beta}_p$ and ℓ_i from external poloidal magnetic measurements alone and therefore provide a useful check of the diamagnetic value.

Statistically, it has been shown that for sufficiently elongated plasmas $\bar{\beta}_p$ can be determined independent of ℓ_i [2] in fitting D-III magnetic data with a free-boundary equilibrium code, which repeatedly solves the non-linear Grad-Shafranov equation to search for the best fitted current density profile. The analytic underpinning of this statistical observation is given in this paper. Integral MHD equilibrium relations are

derived which show that for non-circular plasmas $\bar{\beta}_p$ and ℓ_i can be separately obtained from line integrals of the external magnetic field. The volume-dependent parameters that appear depend only weakly on the actual current density distribution inside the plasma and can be evaluated approximately given the plasma shape and boundary poloidal magnetic field. For a tokamak with a non-circular cross-section of sufficient elongation, $\bar{\beta}_p$ and ℓ_i can then be approximately determined separately. The required plasma shape and boundary poloidal magnetic field are shown to be accurately obtained for most of the diverted and limited plasmas produced in D-III from the measured external poloidal magnetic field and flux values by approximating the plasma current distribution using a few filament currents [3, 7]. The approach here is more general than that given in Ref.[4], where a relationship based on a low-beta ($\bar{\beta}_p$ much less than 1) analytic solution of the moment equation describing the ellipticity is used to provide the separation. The approximation for $\bar{\beta}_p$ obtained in this way is sufficiently accurate and fast to allow on-line data analyses between laboratory plasma discharges. A more physical and accurate, but computationally more extensive, approach where the plasma current is approximated using distributed sources is given in Ref.[8].

In Section 2, integral MHD equilibrium relations relating $\bar{\beta}_p$ and ℓ_i separately to line integrals of the external poloidal magnetic field for non-circular

tokamaks are derived. An analytical example is given in Section 3. In Section 4, comparative calculations and a sensitivity study using D-III data are given.

2. INTEGRAL RELATIONS OF THE ANISOTROPIC STATIONARY MHD EQUILIBRIUM EQUATION

2.1. Derivation

Following Refs [6, 9], a set of integral relations may be obtained from the anisotropic stationary MHD equilibrium equation by multiplying it with an arbitrary vector $\vec{Q}$ and integrating over the volume Ω of a torus containing the plasma:

$$\begin{aligned} \int_{\Omega} dV \left[\left(P_{\perp} + \frac{B_p^2}{8\pi} \right) \nabla \cdot \vec{Q} - (P_{\perp} - P_{\parallel}) \frac{\vec{Q} \cdot \vec{e}_R}{R} \right. \\ \left. + n_m U_t^2 \frac{\vec{Q} \cdot \vec{e}_R}{R} + n_m \vec{U}_p (\vec{U}_p \cdot \nabla) \vec{Q} \right. \\ \left. + \frac{B_t^2 - B_{tv}^2}{8\pi} \left(\nabla \cdot \vec{Q} - \frac{2\vec{Q} \cdot \vec{e}_R}{R} \right) - \frac{\sigma}{4\pi} \vec{B}_p (\vec{B}_p \cdot \nabla) \vec{Q} \right. \\ \left. + (P_{\perp} - P_{\parallel}) \frac{B_p^2}{B^2} \frac{\vec{Q} \cdot \vec{e}_R}{R} \right] \\ = \int_{\Gamma} dS \left[\frac{(B_r^2 - B_n^2)}{8\pi} Q_n - \frac{B_n B_r Q_r}{4\pi} \right] \end{aligned} \quad (1)$$

where $P_{\perp}$ and $P_{\parallel}$ are the perpendicular and the parallel components of the pressure tensor, n_m and $\vec{U}$ are the mass density and flow velocity of the plasma, σ is the anisotropy parameter $\sigma = 1 + 4\pi(P_{\perp} - P_{\parallel})/B^2$, $\vec{B}_p$ and $\vec{B}_t$ are the poloidal and the toroidal components of the magnetic field $\vec{B}$, Γ denotes the surface bounding Ω , $\vec{n}$ and $\vec{\tau}$ are the unit vectors normal and parallel to Γ , B_{tv} is the vacuum toroidal magnetic field, $B_n = \vec{B}_p \cdot \vec{n}$, $B_r = \vec{B}_p \cdot \vec{\tau}$, $Q_n = \vec{Q} \cdot \vec{n}$, $Q_r = \vec{Q} \cdot \vec{\tau}$, and $\vec{U}_t$ and $\vec{U}_p$ are the toroidal and the poloidal components of $\vec{U}$. Axisymmetry in a R, Z co-ordinate system is assumed; $\vec{Q}$ is taken to have no toroidal component; and $P_{\perp}$, $P_{\parallel}$, and $\vec{U}$ are assumed to vanish on Γ .

The integral relations given in Eq.(1) represent the generalizations of the ones derived originally by

Zakharov and Shafranov for a static isotropic plasma [9] to include pressure anisotropy and plasma rotation. Three particular choices of $\vec{Q}$, $R\vec{e}_R + Z\vec{e}_Z$, $\vec{e}_R$, and $Z\vec{e}_Z$, are especially useful and yield

$$2\bar{\beta}_{p\perp} + \bar{\beta}_{p\parallel} + 2W_{pt} + \ell_i - \mu_i = S_1 + S_2 \quad (2)$$

$$\bar{\beta}_{p\parallel} + 2W_{pt} + \ell_i + \mu_i = \frac{R_t}{R_0} S_2 \quad (3)$$

$$\bar{\beta}_{p\perp} - (\alpha - 1) \ell_i - \mu_i = S_3 \quad (4)$$

Here, $\bar{\beta}_{p\perp}$, $\bar{\beta}_{p\parallel}$, and W_{pt} are the components of poloidal plasma beta associated with the perpendicular pressure, the parallel pressure, and the flow, respectively, while ℓ_i is the internal inductance of the plasma, and μ_i is the plasma diamagnetism. These are defined by

$$\begin{aligned} [\bar{\beta}_{p\perp}, \bar{\beta}_{p\parallel}, W_{pt}, \ell_i, \mu_i] = \frac{8\pi}{B_{pa}^2 \Omega} \\ \times \int_{\Omega} dV \left[P_{\perp}, p_{\parallel}, \frac{1}{2} n_m U_t^2, \frac{B_p^2}{8\pi}, \frac{B_{tv}^2 - B_t^2}{8\pi} \right] \end{aligned} \quad (5)$$

The remaining quantities are

$$S_1 = \frac{1}{B_{pa}^2 \Omega} \int_{\Gamma} dS B_p^2 (R\vec{e}_R + Z\vec{e}_Z - R_0\vec{e}_R) \cdot \vec{n} \quad (6)$$

$$S_2 = \frac{1}{B_{pa}^2 \Omega} \int_{\Gamma} dS B_p^2 R_0 \vec{e}_R \cdot \vec{n} \quad (7)$$

$$S_3 = \frac{1}{B_{pa}^2 \Omega} \int_{\Gamma} dS B_p^2 Z \vec{e}_Z \cdot \vec{n} \quad (8)$$

$$R_t = \int_{\Omega} dV g / \int_{\Omega} dV \frac{g}{R} \quad (9)$$

$$g = P_{\parallel} + n_m U_t^2 + \frac{B_p^2 + B_{tv}^2 - B_t^2}{8\pi} \quad (10)$$

$$\alpha = 2 \int_{\Omega} dV (\vec{B}_p \cdot \vec{e}_Z)^2 / \int_{\Omega} dV B_p^2 \quad (11)$$

with R_0 and B_{pa} a constant major radius and poloidal magnetic field for normalization. A convenient choice

for B_{pa} based on Ampère's law is given by

$$B_{A0} = \oint_{\Gamma} d\ell B_p / \oint_{\Gamma} d\ell$$

For all calculations in this paper, R_0 is chosen to have the value of 1.43 m.

In arriving at Eqs (2) to (4), the small last term on the left-hand side of Eq.(1) associated with the pressure anisotropy as well as the poloidal rotation flow terms have been neglected, and Ω has been taken to be the plasma volume. In Eqs (2) to (4), there are five unknowns $\bar{\beta}_{p\perp}$, $\bar{\beta}_{p\parallel}$, ℓ_i , μ_i , and W_{pt} , three surface-dependent parameters, S_1 , S_2 , and S_3 , and two volume-dependent parameters α and R_t . Approximate evaluations of α and R_t in terms of the surface parameters are given in the next section.

Equations (2) and (3) were given by Wootton and Cooper [6] and were originally derived for the isotropic static case by Shafranov [5]. They have been fully exploited in the past and can be combined to yield

$$\frac{1}{2}(\bar{\beta}_{p\perp} + \bar{\beta}_{p\parallel}) + W_{pt} + \frac{\ell_i}{2} = \frac{S_1}{4} + \frac{S_2}{4} \left(1 + \frac{R_t}{R_0}\right) \quad (12)$$

Equation (4) is the new relation which will be shown in the later sections to provide a satisfactory separation of $\bar{\beta}_p$ and ℓ_i for non-circular plasmas.

A useful relation relating $\bar{\beta}_{p\perp}$ to the diamagnetism may be obtained from Eqs (2) and (3) by eliminating ℓ_i :

$$\bar{\beta}_{p\perp} = \frac{S_1}{2} + \frac{S_2}{2} \left(1 - \frac{R_t}{R_0}\right) + \mu_i \quad (13)$$

The diamagnetism μ_i may be approximately related to the diamagnetic flux $\Delta\phi$ through

$$\mu_i \cong \frac{4\pi B_{t0} R_0 \Delta\phi}{B_{pa}^2 \Omega} \quad (14)$$

where B_{t0} is the vacuum toroidal magnetic field at R_0 and

$$\Delta\phi = - \int_{\Omega} dS_t (B_t - B_{t0})$$

Here, $dS_t = dV/(2\pi R)$ is the cross-sectional area of Ω . Neglecting the second small term and approximating S_1 and Ω by using elliptical flux surface geometry in Eq.(13) yield the convenient formula

$$\bar{\beta}_{p\perp} = 1 + \frac{E^2 + 1}{2E} \frac{B_{t0} \Delta\phi}{20\pi I_0^2} \quad (15)$$

where E is the plasma ellipticity, B_{t0} is the vacuum toroidal magnetic field at the plasma centre in tesla, $\Delta\phi$ is the diamagnetic flux in millivolt-seconds, and I_0 is the plasma current in mega-amperes. Equation (15) has been used extensively to interpret $\bar{\beta}_{p\perp}$ from diamagnetic measurements. Note that the poloidal beta as determined from diamagnetic measurements using Eq.(15) is $\bar{\beta}_{p\perp}$, whereas the one determined from magnetic measurements using Eq.(12) is $(\bar{\beta}_{p\perp} + \bar{\beta}_{p\parallel})/2 + W_{pt}$. Thus, for shaped tokamaks, magnetic together with diamagnetic measurements may be used to determine the plasma anisotropy and toroidal flow parameter $(\bar{\beta}_{p\parallel} - \bar{\beta}_{p\perp})/2 + W_{pt}$. The plasma anisotropy $\bar{\beta}_{p\parallel} - \bar{\beta}_{p\perp}$ may then be determined by estimating the poloidal beta associated with toroidal rotation W_{pt} from spectroscopic measurements.

2.2. Approximate evaluations of volume-dependent parameters

In this section, the two volume-dependent parameters appearing in Eqs (2) to (4) are evaluated approximately in terms of the surface parameters. They are shown to depend only weakly on the current density distribution inside the plasma. The parameter α characterizes the ratio between twice the magnetic energy stored in the vertical poloidal field and the poloidal magnetic energy stored in the plasma. For plasmas with magnetic surfaces close to elliptical, α can be shown to be approximately independent of the current density distribution inside the plasma and to depend on the plasma elongation E only:

$$\alpha \cong \frac{2E^2}{E^2 + 1} + 0 \left(\frac{1}{A^2} \right) \quad (16)$$

where A is the aspect ratio of the plasma. The details of the derivation are given in Appendix A.

Since B_p has the largest value at the boundary, another approximation to α may be obtained by evaluating the integrals for B_z and B_p over a thin shell around the boundary $\Delta\Omega$:

$$\alpha \cong 2 \int_{\Delta\Omega} dV (\vec{B}_p \cdot \vec{e}_z)^2 / \int_{\Delta\Omega} dV B_p^2 \quad (17)$$

Equation (17) provides a better approximation to α than Eq.(16) if the plasma has a rapid variation of ellipticity near the boundary and very peaked toroidal current density profile such as in the diverted plasmas produced in D-III. The required B_z and B_p may be

obtained by using the filament plasma current model [3, 7] as discussed in the latter sections.

For a straight circular cylindrical plasma, $\alpha = 1$, $R_t/R_0 = 1$, $S_3 = S_1/2 = 1$, Eq.(4) becomes a linear combination of Eqs (2) and (3) and no longer yields any useful information on ℓ_i or $\bar{\beta}_p$. This is consistent with the physical fact that for a straight circular cylinder, external poloidal magnetic measurements alone yield information on only the total plasma current, not on the current density itself. On the other hand, as the elongation E is increased, Eq.(4) becomes less dependent on Eqs (2) and (3), and more accurate separate determinations of $\bar{\beta}_p$ and ℓ_i become possible. This is consistent with the statistical picture given in Ref.[2] that with significant non-circularity ℓ_i can be determined independent of $\bar{\beta}_p$ from magnetic measurements alone.

Assuming $\alpha \neq 1$, Eqs (2)–(4) can be combined to yield

$$\ell_i = \frac{1}{\alpha - 1} \left[\frac{S_1}{2} + \frac{S_2}{2} \left(1 - \frac{R_t}{R_0} \right) - S_3 \right] \quad (18)$$

Because of the near cancellation between $S_1/2$ and S_3 , ℓ_i is more sensitive to the value of the volume-dependent parameter R_t used in its evaluation than is the sum

$$(\bar{\beta}_{p\perp} + \bar{\beta}_{p\parallel})/2 + W_{pt} + \ell_i/2$$

Several approximations to R_t are possible. One approximation may be obtained from Eq.(9) by setting

$$R_t \cong R_A = \int_{\Omega} dV / \int_{\Omega} \frac{dV}{R} = \frac{\Omega}{2\pi S_t} \quad (19)$$

where S_t is the cross-sectional area of Ω . Another approximation to R_t is given by the plasma current centroid

$$\begin{aligned} R_t^2 &\cong R_c^2 = \int_{\Omega} dR dZ J_t R^2 / I_0 \\ &= c \oint_{\Gamma} d\ell B_p R^2 / 4\pi I_0 \end{aligned} \quad (20)$$

where J_t is the toroidal current density.

As will be shown in the latter sections, these approximations to α and R_t provide satisfactory separation of $\bar{\beta}_p$ and ℓ_i for sufficiently elongated plasmas. A more accurate, although computationally more extensive, approximation to α and R_t may be

TABLE I. COMPARISONS OF α AND R_t FOR FOUR EQUILIBRIA WITH APPROXIMATELY SIMILAR BOUNDARY SHAPES AND POLOIDAL MAGNETIC FIELD

	Case a	Case b	Case c	Case d
$R_t(\text{cm})$	142.5	142.5	142.6	142.3
α	1.38	1.38	1.37	1.39
q_m	0.95	1.18	0.50	2.00
q_l	2.46	2.47	2.43	2.51
$\Omega (\text{m}^3)$	6.87	6.88	6.80	6.94
$\bar{\beta}_p$	0.91	0.91	0.90	0.92
$\bar{\beta}_t(\%)$	4.21	4.21	4.18	4.20
ℓ_i	0.76	0.76	0.78	0.75
E	1.60	1.59	1.59	1.60
$S_t(\text{m}^2)$	0.78	0.78	0.77	0.79
S_1	2.03	2.03	2.02	2.04
S_2	1.52	1.52	1.49	1.55
S_3	0.73	0.72	0.72	0.73

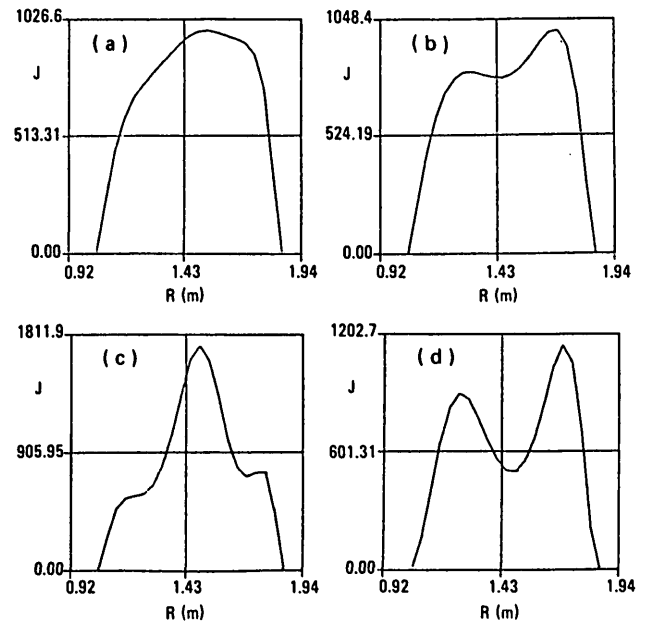


FIG.1. Toroidal current density distributions in horizontal plane passing through magnetic axis for four equilibria of approximately the same plasma shape and boundary poloidal magnetic field.

obtained by modelling the plasma current density distribution more physically using distributed sources that best fit the externally measured magnetic data and thus allow a more accurate separation of $\bar{\beta}_p$ and ℓ_i for non-circular plasmas [8].

The insensitivity of α and R_t to the toroidal current density distribution inside the plasma is illustrated in Table I, where the values of R_t and α computed exactly from the given current distributions using Eqs (9) and (11) are compared for four limited D-shaped equilibria which have different current distributions (Fig.1) but approximately the same shape and boundary poloidal magnetic field. Also shown are the safety factor at the magnetic axis and at the boundary q_m and q_l . Although the current density distributions are vastly different, α and R_t have about the same values. The approximate values of α and R_t as given by Eqs (16) and (19) are about 1.43 cm and 140 cm, respectively, which agree well with the values computed exactly.

3. AN ANALYTICAL EXAMPLE

As an illustration of the uses of the integral relations, in this section a particular analytic Solovév equilibrium [10] is considered. The normalized poloidal flux $\hat{\psi} = \psi/(2B_0\pi a^2)$ varying between 0 and 1 is given by

$$\hat{\psi} = \frac{\beta_{t0}}{8}(r^2 - r_m^2)^2 + \alpha_F r_m^2 z^2 \quad (21)$$

The normalized pressure $\hat{P} = 4\pi P/B_0^2$ and the square of the normalized poloidal current $\hat{F}^2 = 2F/(cB_0 r_m a)$ are linear in $\hat{\psi}$,

$$\hat{P}(\hat{\psi}) = \beta_{t0}(1 - \hat{\psi}), \quad \hat{F}^2(\hat{\psi}) = 1 - 4\alpha_F \hat{\psi}$$

The toroidal current density profile is flat. Here, $r = R/a$, $z = Z/a$, $r_m = 4.0$ is the aspect ratio of the magnetic axis, $\beta_{t0} = 1/8$, $\alpha_F = 1/40$, and a and B_0 are some constant length and magnetic field for normalization. The normalized plasma energy W_p and the normalized poloidal magnetic energy W_B can be calculated to be

$$\begin{aligned} W_p &= \frac{4\pi}{a^3 B_0^2} \int_{\Omega} dV P \\ &= \frac{\pi^2}{r_m} \left(\frac{2\beta_{t0}}{\alpha_F} \right)^{1/2} \end{aligned} \quad (22)$$

$$\begin{aligned} W_B &= \frac{4\pi}{a^3 B_0^2} \int_{\Omega} dV \frac{B_p^2}{8\pi} \\ &= \pi^2 \tilde{E} \left(\frac{\beta_{t0}}{2} \right)^{1/2} \left\{ 1 + \frac{r_m^2}{\tilde{E}} \left[1 - \frac{1}{6C_1^2} \left(1 - [1 - 4C_1^2]^{3/2} \right) \right] \right\} \end{aligned} \quad (23)$$

where

$$\tilde{E} = 1/(r_m \sqrt{\alpha_F}) = \sqrt{10}/2$$

characterizes the plasma elongation and

$$C_1 = (2/\beta_{t0})^{1/2}/r_m^2$$

The details of the calculations are given in Appendix B. Using the choice of B_{pa} based on Ampère's law B_{A0} , $\bar{\beta}_p$ and ℓ_i can be computed analytically from Eqs (22) and (23) to have the exact values of 0.682 and 0.484, respectively. Similarly, the two volume-dependent parameters R_t and α can be computed from Eq.(9) and Eqs (B.6) and (B.7) to have the exact values of 3.84 and 1.41, respectively.

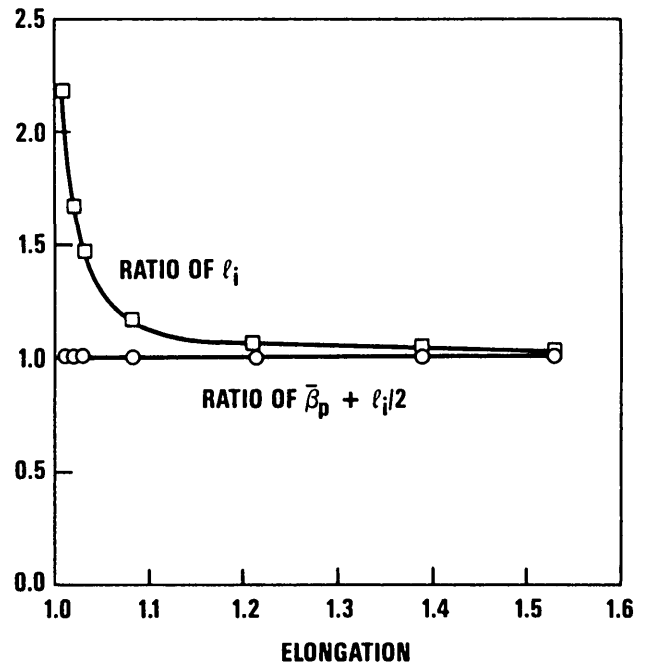


FIG.2. Ratios of approximate ℓ_i and $\bar{\beta}_p + \ell_i/2$ to their exact values as a function of elongation E for a sequence of Solovév equilibria.

On the other hand, the approximate value of α as given by Eq.(16) is 1.40 and R_t as given by Eqs (19) and (20) are 3.90 and 3.96, respectively. Using these approximate values of α and R_t and the boundary poloidal magnetic field as given by Eq.(21), the approximate values of $\bar{\beta}_p$ and ℓ_i can be computed by using the integral relations given in Eqs (12) and (18) to be 0.677 and 0.495 for R_t approximated by using Eq.(19) and 0.689 and 0.478 for R_t approximated by using Eq.(20). In either case, these values of $\bar{\beta}_p$ and ℓ_i agree well with the exact values.

In Fig.2, the ratios of the approximate values of ℓ_i and $\bar{\beta}_p + \ell_i/2$, computed using α and R_t as given by Eqs (16) and (19), to their exact values are shown for a sequence of Solovév equilibria with varying elongation E . Here, E is varied by changing α_F . As the plasma shape becomes more circular, $\bar{\beta}_p$ and ℓ_i are no longer separable for $E \leq 1.1$. In practice, because of noise in the magnetic signals, higher elongations are required.

4. COMPARATIVE CALCULATIONS

4.1. Determination of plasma shape and external magnetic field

Generally, the plasma shape and the boundary magnetic field required for evaluations of approximate values of $\bar{\beta}_p$ and ℓ_i are not directly measurable but must be reconstructed from the poloidal magnetic field and flux measurements on some fixed locations surrounding the plasma. To determine the required plasma shape and the boundary magnetic field from these measurements, an efficient technique originally developed by Swain and Neilson [3] is used here. It may be viewed as an efficient extrapolation scheme to get from the measured values to the needed values at the plasma boundary. The method has been described in detail in Ref.[3] and is only briefly summarized here.

An essential feature of the technique is the approximation of the plasma toroidal current density J_t by using a few filaments at pre-specified locations. In essence, this is equivalent to doing a multipole expansion of the plasma current and truncating the higher-order moments. The method is similar to that given in Ref.[7], which uses a surface plasma current approximation. Unlike Ref.[3], one difference here is that the measured values of the poloidal magnetic fluxes at the external shaping coils rather than the currents are used for the reconstruction for two reasons. First, the

fluxes can be measured more accurately than the currents. Secondly, the eddy currents flowing in the vacuum chamber can be somewhat accounted for by requiring the fluxes at the shaping coils rather than the currents be matched in the reconstruction. The unknown plasma filament and the external coil currents $I_j, j = 1, n$, are then determined by minimizing the sum of the squares of the differences, χ^2 , between the values of the measured poloidal magnetic fields and fluxes $M_i, i = 1, m$, and the computed values C_i :

$$\chi^2 = \sum_{i=1}^m \left(\frac{M_i - C_i}{\sigma_i} \right)^2 + \zeta \sum_{j=1}^n I_j^2 \quad (24)$$

where σ_i is the standard deviation reflecting the uncertainty in the measurement of the i field quantity. Here, ζ is a small constant and the last term has been added for numerical stabilization. Setting $\partial\chi^2/\partial I_j = 0$ yields a set of linear equations for determining I_j

$$\vec{H} \cdot \vec{I} = \vec{b} \quad (25)$$

Here, the unknown vector $\vec{I}$ contains the plasma filament and the external coil currents, the symmetric response matrix $\vec{H}$ and the vector $\vec{b}$ are given by

$$(\vec{H})_{ij} = \sum_{k=1}^m \frac{G_{ki} G_{kj}}{\sigma_k^2} + 2\zeta \delta_{ij} \quad (26)$$

$$(\vec{b})_i = \sum_{k=1}^m \frac{G_{ki} M_k}{\sigma_k^2} \quad (27)$$

δ_{ij} is the Dirac delta function, and G_{kj} is the response function which linearly relates the k -th field quantity to the j -th current

$$C_k = \sum_{j=1}^n G_{kj} I_j \quad (28)$$

Once the currents have been determined, the plasma boundary can be found by searching for the largest closed flux surface enclosed by the limiter contour. The diversity of plasma shapes attainable in D-III requires the use of an elaborate iteration scheme to obtain the boundary. In a single discharge, the plasma can evolve from a circular limited shape to a highly elongated and triangular diverted configuration. Thus, simple techniques, such as that based on perturbing a reference contour, are not applicable.

An algorithm which is capable of handling both limited and diverted discharges has been developed and is based on a quasi-linear interpolation of the poloidal flux and an iteration proceeding from the plasma centre outward to the limiter contour until the largest closed flux surface is found. Given two points $\bar{x}_1$, $\bar{x}_2$ such that $\bar{x}_1$ is inside and $\bar{x}_2$ outside the plasma, the line $\bar{x}_1 + \lambda(\bar{x}_2 - \bar{x}_1)$ is searched for the smallest value of the poloidal flux function, ψ , subject to the constraint that ψ yields a closed flux surface bounded on all sides by the limiter and the vacuum chamber walls. For a limited plasma this procedure automatically yields the flux surface which is tangent to the limiter or the vacuum chamber wall. For a diverted plasma all flux contours on or outside the separatrix intersect the limiter wall or the vacuum chamber at some point and are thus rejected. Consequently, this method does not yield the poloidal flux value ψ_s at the separatrix exactly. Rather, it determines a value of ψ , ψ_1 , such that $|\psi_1 - \psi_s|$ less than some tolerance, where ψ_1 defines a closed contour just inside the separatrix. The error $\psi_1 - \psi_s$ is governed by the grid density on which ψ is known and the contour tracing scheme used to determine whether or not a flux contour is closed. This leads to an approximation for the x point which is sufficiently accurate in many cases. The exact location of the x point can then be found by using this approximation as an initial guess for the Newton-Raphson method, used to solve the simultaneous equations $\partial\psi/\partial R = 0$ and $\partial\psi/\partial Z = 0$. The differences between ψ on the separatrix found from these equations and ψ_1 have always been found to be $|\psi_1 - \psi_s|/|\psi_m - \psi_s| < 0.03$, where ψ_m is the poloidal flux at the magnetic axis. Hence, the co-ordinates associated with ψ_1 are an adequate representation of the true contour ψ_s . In the vicinity of the x point, where $\nabla\psi$ is small, the co-ordinates of ψ_1 must be replaced with those of ψ_s . This is accomplished using the Taylor series representation of ψ about the x point. In practice, it has been found that only the single point of ψ_1 closest to the true x point must be replaced.

Once the plasma boundary and the poloidal magnetic field along the boundary have been determined, the three line integrals S_1 , S_2 , and S_3 and the approximate values of the two volume-dependent parameters α and R_t can be computed. $\bar{\beta}_p = \frac{1}{2}(\bar{\beta}_{p1} + \bar{\beta}_{p2}) + W_{pt}$ and ℓ_i are then followed from Eqs (12) and (18).

Finally, note that in order for the problem to be well posed, the number of independent magnetic data m must exceed or equal the sum of the number of the filament and external coil currents n .

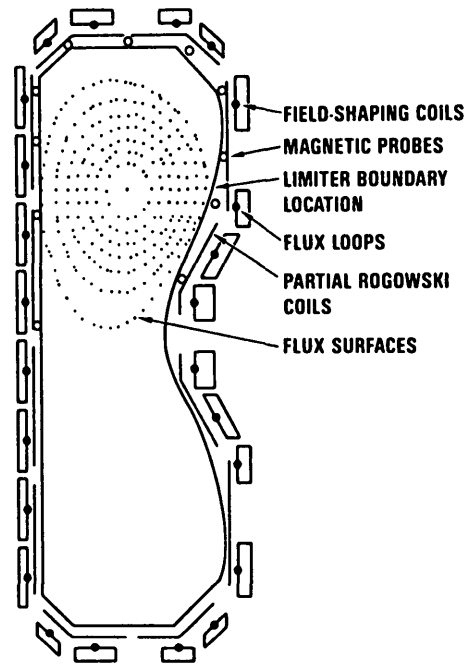


FIG. 3. Magnetic diagnostics of D-III.

4.2. Comparisons

In this section, to assess the applicability of this method in separating $\bar{\beta}_p$ and ℓ_i , two types of comparative calculations using the D-III configuration are performed. The magnetic diagnostics of D-III used in the calculations are shown in Fig. 3. They consist of 24 flux loops, 12 partial Rogowski loops, 11 magnetic probes, and 1 full Rogowski loop. There are 24 external poloidal shaping coils. The plasma current distribution is modelled by using six filament currents distributed around an ellipse inside the plasma.

4.2.1. Comparison against given current density distributions

In this first type of comparison, from a given set of toroidal current density distributions, magnetic signals are simulated for the D-III diagnostics by using a standard two-dimensional MHD equilibrium code. From these signals, the plasma shape, line integrals of the boundary poloidal magnetic field, $\bar{\beta}_p$ and ℓ_i are computed as described above and are compared to the initial known values from the equilibrium calculations.

In Figs 4(a)–4(d), the plasma boundary shapes reconstructed using the filament-current approximation

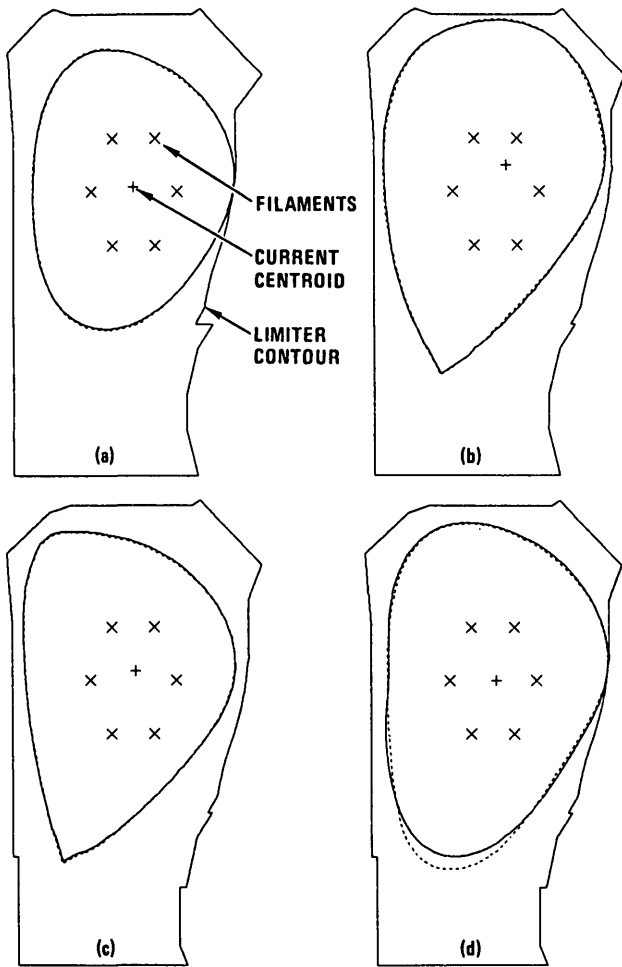


FIG. 4. Comparisons of D-III plasma shapes. Solid contours are computed by using filament-current model, and dashed contours are initial equilibrium results.

are compared to the initial results from the equilibrium calculations for four particular types of plasmas produced in D-III: a medium-size D-shaped plasma, a round-top diverted plasma, a triangular-top diverted plasma, and a large-size D-shaped plasma. The solid contours are computed by using the filament-current model, and the dashed contours represent the initial results from the equilibrium calculations. Also shown are the envelopes of the limiters and the positions of the filaments. The axial safety factor, q_m , is constrained to have the value of 0.95 in the initial equilibrium calculations. The two limited cases have broad toroidal current density profiles, and the two diverted cases have relatively peaked current profiles.

The minor radius $a = (R_{\max} - R_{\min})/2$, the elongation $E = (Z_{\max} - Z_{\min}) / (R_{\max} - R_{\min})$, the major radius and the vertical position of the geometric centre $R_g = (R_{\max} + R_{\min})/2$ and $Z_g = (Z_{\max} + Z_{\min})/2$, the plasma volume Ω , the toroidal current I_0 , the sum $\beta_p + \ell_i/2$, and the line integrals S_1 , S_2 , and S_3 computed

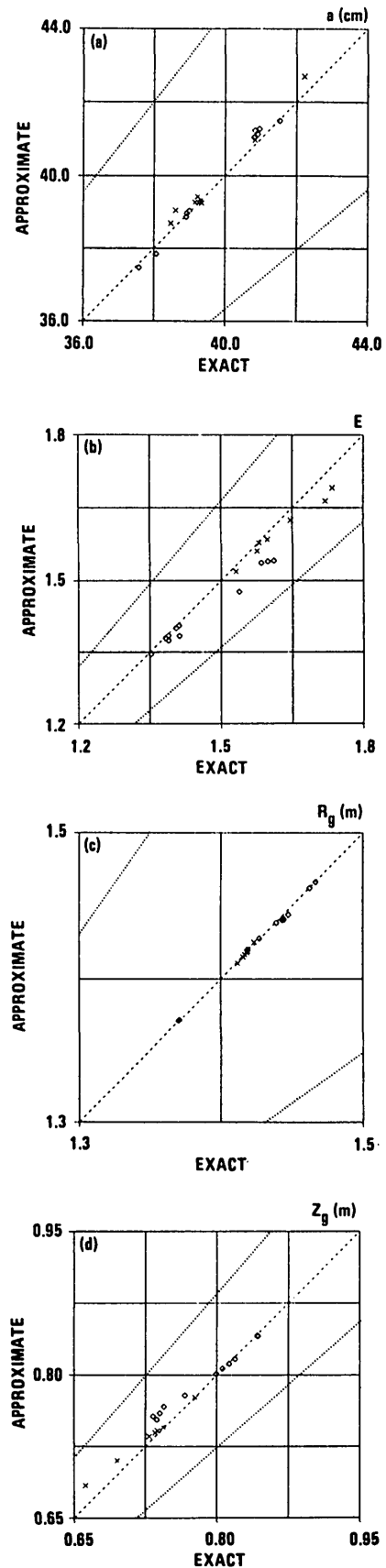
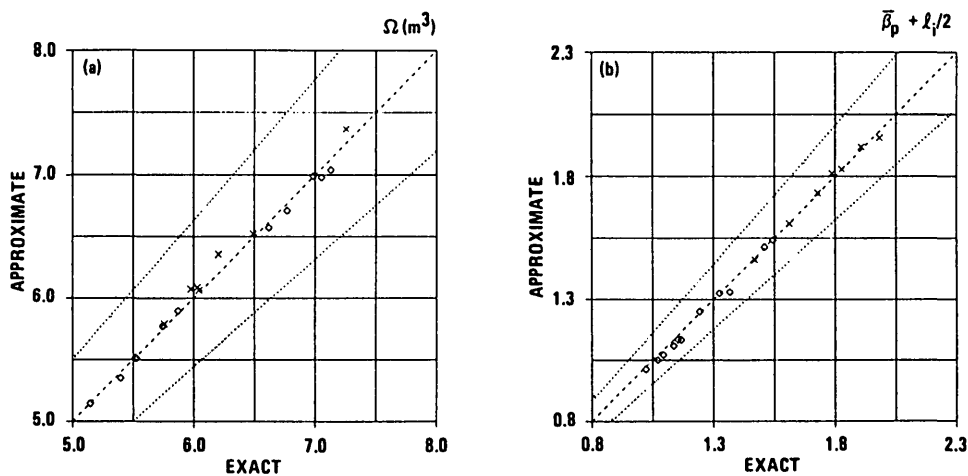


FIG. 5. Comparisons of a , E , R_g , and Z_g . $\diamond$ and $\times$ denote limited and diverted plasmas, respectively. Dotted lines are $\pm 10\%$ lines.

TABLE II. COMPARISONS OF VARIOUS PARAMETERS FOR FOUR DIFFERENT TYPES OF D-III PLASMAS

	Case a		Case b		Case c		Case d	
	Approximate	Given	Approximate	Given	Approximate	Given	Approximate	Given
a (cm)	37.5	37.6	41.0	40.8	39.4	39.2	41.0	40.8
E	1.35	1.35	1.56	1.58	1.52	1.53	1.48	1.54
R _g (cm)	143	143	143	142	142	142	144	145
Z _g (cm)	81	81	78	78	74	74	77	74
Ω (m ³)	5.15	5.15	6.52	6.49	5.79	5.76	6.71	6.77
I ₀ (kA)	345	345	420	419	328	328	507	510
$\bar{\beta}_p + \ell_i/2$	1.01	1.02	1.92	1.91	1.83	1.83	1.14	1.17
$\bar{\beta}_t$ (%)	4.38	4.36	2.24	2.16	1.27	1.25	3.80	4.03
ℓ_i	0.65	0.68	0.98	1.03	1.16	1.21	0.78	0.71
$\bar{\beta}_t^*$ (%)	4.32	4.36	—	—	—	—	3.82	4.03
ℓ_i^* (%)	0.67	0.68	—	—	—	—	0.76	0.71
S ₁	1.978	1.954	2.294	2.274	2.321	2.318	1.980	1.978
S ₂	1.039	1.018	2.688	2.613	2.520	2.508	1.286	1.256
S ₃	0.807	0.800	0.839	0.835	0.886	0.890	0.721	0.744
q _m	—	0.95	—	0.95	—	0.95	—	0.95
q _t /q [*]	1.51	1.52	4.33	4.40	4.92	4.93	2.18	2.31
S _t (m ²)	0.585	0.585	0.745	0.741	0.673	0.669	0.762	0.769
R _A /R _t (cm)	140.0	137.8	139.4	141.6	136.9	139.3	140.2	138.7
α^*/α	1.289	1.286	—	1.295	—	1.233	1.371	1.358

FIG. 6. Comparisons of Ω and $\bar{\beta}_p + \ell_i/2$. ◇ and × denote limited and diverted plasmas, respectively. Dotted lines are $\pm 10\%$ lines.

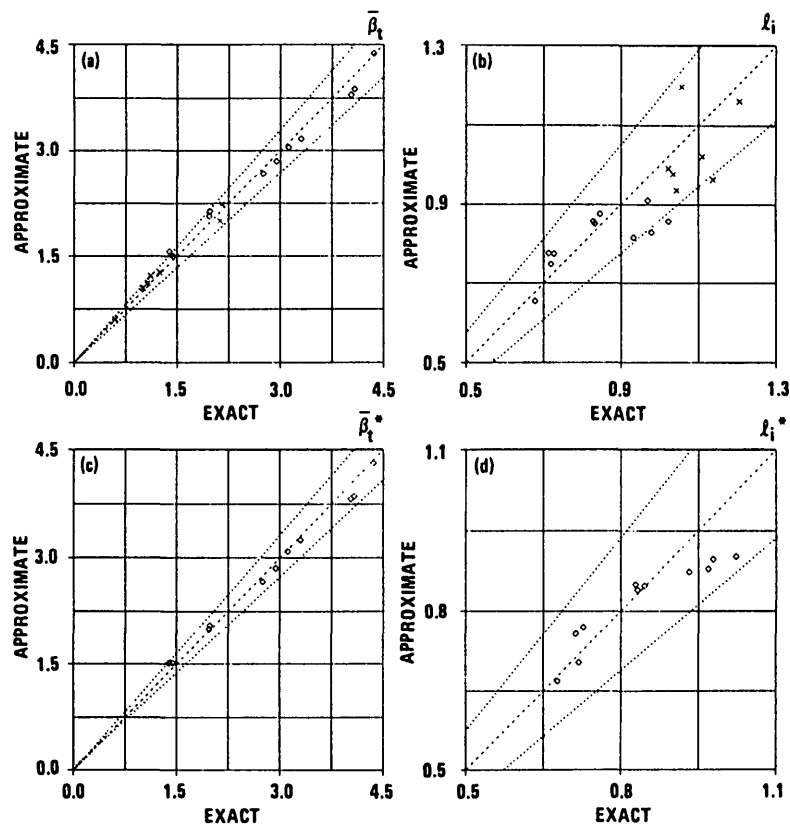


FIG. 7. Comparisons of β_t , l_i , β_t^* , and l_i^* . $\diamond$ and $\times$ denote limited and diverted plasmas, respectively. Dotted lines for β_t and β_t^* are $\pm 10\%$ lines whereas those for l_i and l_i^* are $\pm 15\%$ lines.

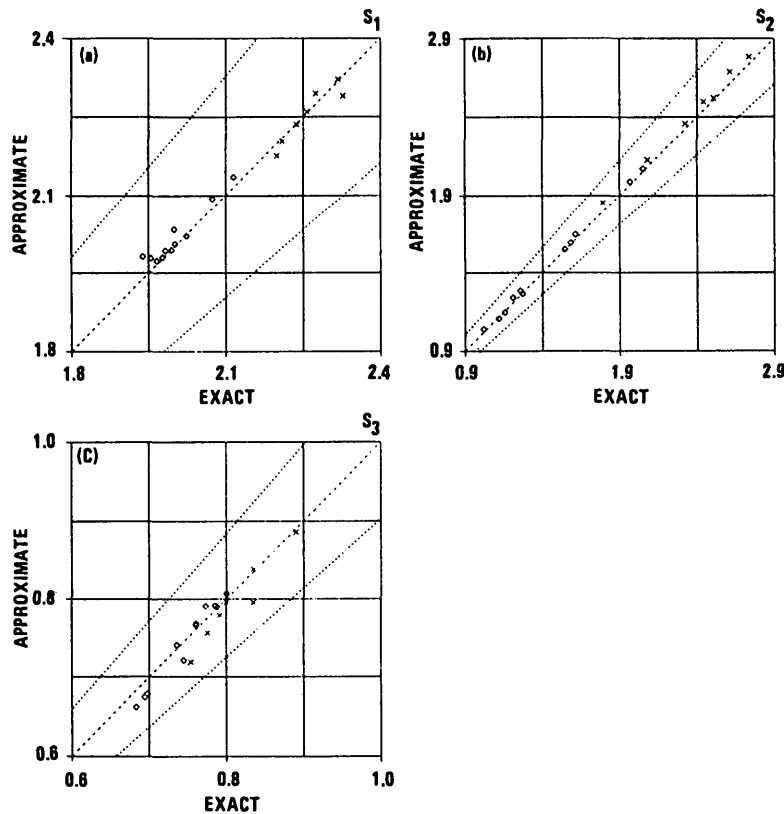


FIG. 8. Comparisons of S_1 , S_2 , and S_3 . $\diamond$ and $\times$ denote limited and diverted plasmas, respectively. Dotted lines are $\pm 10\%$ lines.

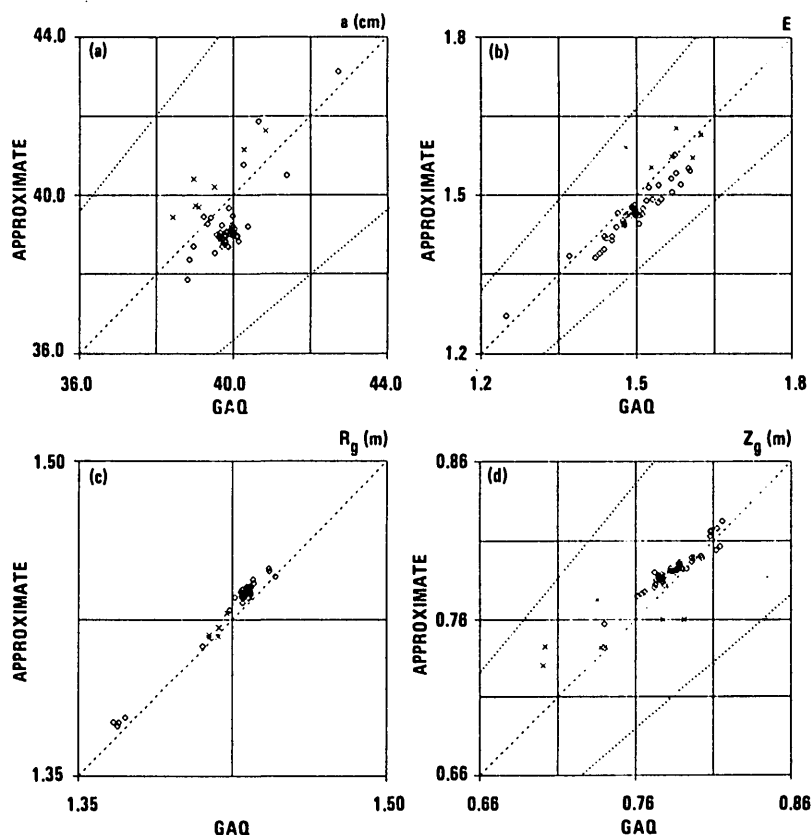


FIG. 9. Comparisons of a , E , R_g , and Z_g . $\diamond$ and $\times$ denote limited and diverted plasmas, respectively. Dotted lines are $\pm 10\%$ lines.

using this approximate method are compared to the initial equilibrium values in Table II for the four cases. Here, $R_{\max}$, $Z_{\max}$, $R_{\min}$, and $Z_{\min}$ are the maximum and minimum of R and Z at the boundary. Also compared are the average toroidal beta $\bar{\beta}_t = \bar{\beta}_p B_{pa}^2 / B_t^2(R_g)$ and the internal inductance ℓ_i as computed using the approximate value of α given by Eq.(17) and R_t approximated by $(R_A + R_C)/2$, the safety factor at the boundary q_l for the limited plasmas, and the equivalent boundary safety factor q^* [8] for the diverted plasmas. The average toroidal beta $\bar{\beta}_t^*$ and the internal inductance ℓ_i^* for the two limited cases computed using the approximate value of α given by Eq.(16), α^* , and R_t approximated by R_A are also compared. In all cases, the agreement is good.

Finally, in Figs 5 to 8, the approximate values of these parameters computed as described above are compared to the initial values from the equilibrium calculations for a variety of plasmas with elongation E varying between 1.3–1.7, $\bar{\beta}_t$ between 0.5–4.4% and ℓ_i between 0.6 and 1.25. The $\diamond$ and the $\times$ symbols denote limited and diverted plasmas, respectively. Again, the toroidal current density profiles are broad in the limited cases and relatively peaked in the

diverted cases. The axial safety factors q_m are constrained to be 0.95 in the initial equilibrium calculations. The agreement for the plasma shape parameters and the sum $\bar{\beta}_p + \ell_i/2$ is excellent. On the other hand, $\bar{\beta}_t$ and $\bar{\beta}_t^*$ agree to within 10% with the initial equilibrium values, whereas ℓ_i and ℓ_i^* agree to within 15%.

4.2.2. Comparison against other methods

In this second type of comparison, the actual measured signals from the magnetic diagnostics for a number of limited and diverted D-III discharges over a wide range of elongation E are used to determine the plasma shapes, the required line integrals of the boundary poloidal magnetic field, and the approximate values of beta using the approximate method described above. The results are then compared to those determined using the GAQ equilibrium code based on the approach given by Luxon and Brown as described earlier [2] and the diamagnetic measurements. The effects of pressure anisotropy and toroidal rotation are small and are neglected in this comparison. In Figs 9 and 10, the approximate values of a , E , (R_g, Z_g) , Ω , $\bar{\beta}_p + \ell_i/2$, $\bar{\beta}_t$ and $\bar{\beta}_t^*$ are compared against those deter-

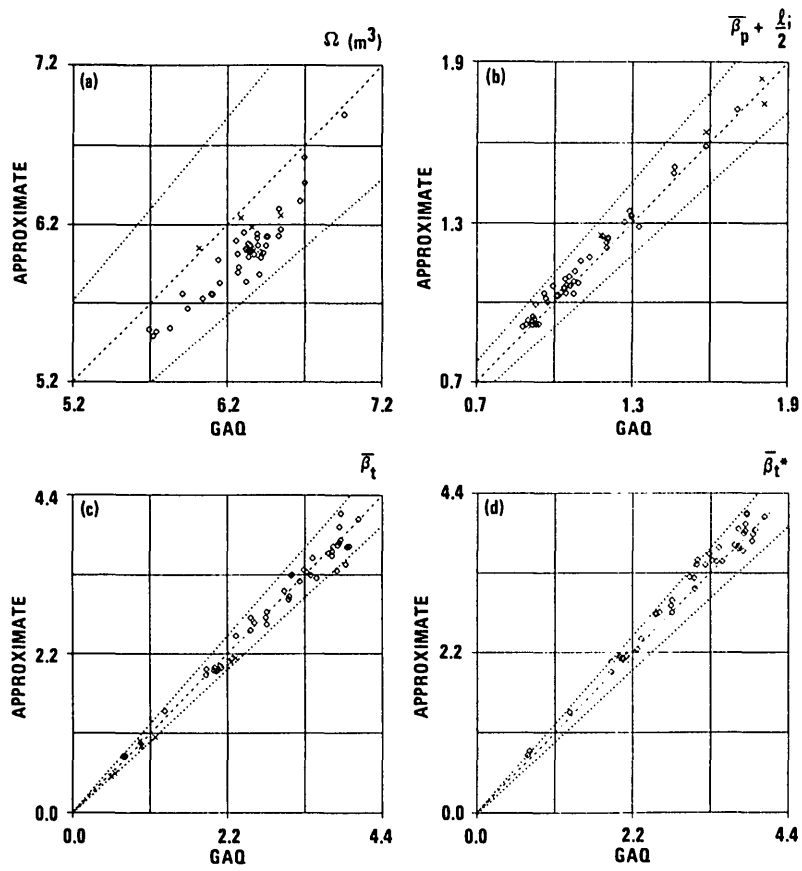


FIG.10. Comparisons of Ω , $\beta_p + l_i/2$, β_t , and β_t^* . ◇ and × denote limited and diverted plasmas, respectively. Dotted lines are $\pm 10\%$ lines.

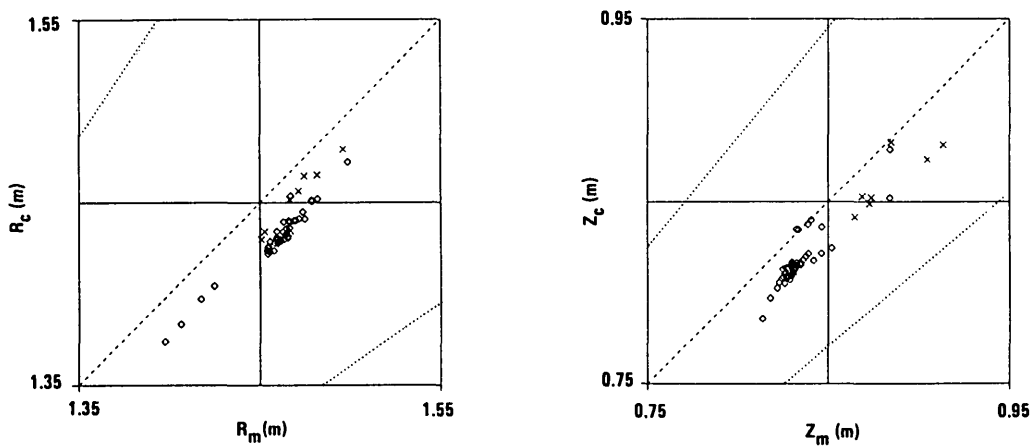


FIG.11. Comparison of position of current centroid (R_c , Z_c) against magnetic axis (R_m , Z_m). ◇ and × denote limited and diverted plasmas, respectively. Dotted lines are $\pm 10\%$ lines.

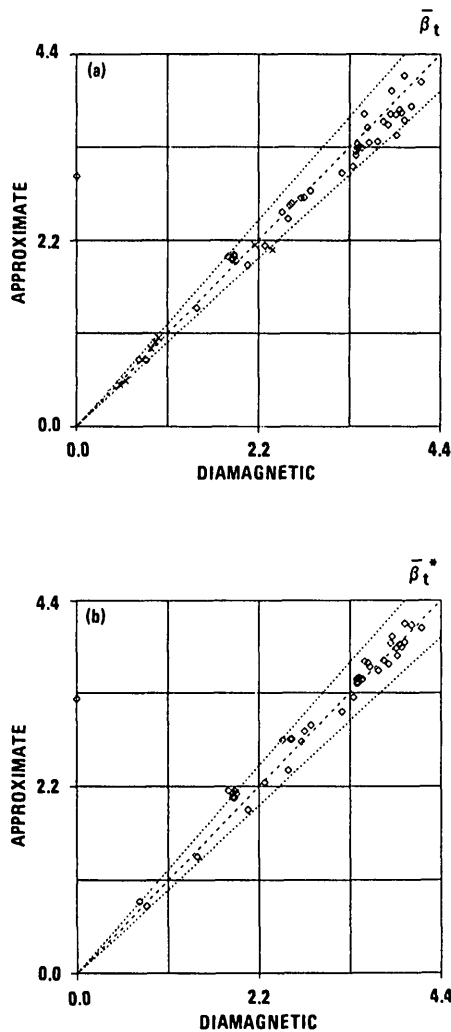


FIG. 12. Comparisons of $\bar{\beta}_t$ and $\bar{\beta}_t^*$ against diamagnetic values. $\diamond$ and $\times$ denote limited and diverted plasmas, respectively. Dotted lines are $\pm 10\%$ lines.

mined using the GAQ code. Again, the $\diamond$ and $\times$ symbols denote limited and diverted D-III discharges, respectively. In Fig. 11, the positions of the current centroid (R_c , Z_c) from this approximate method are compared against the positions of the magnetic axis (R_m , Z_m) from GAQ. Finally, in Fig. 12 the values of $\bar{\beta}_t$ and $\bar{\beta}_t^*$ computed using this approximate method are compared against the values from the diamagnetic measurements. In all cases, the agreement is good.

4.3. Sensitivity study

In this section, the effects of measurement errors on the reconstructed plasma shapes, beta values, and ℓ_i

are studied. To simulate the measurement errors in actual experimental situations, the simulated signals at the measuring probes from the equilibrium calculations are randomly perturbed from -3% to 3% , an uncertainty level appropriate for D-III. The results of the fitting calculations computed using the approximate method described above for the limited D-shaped plasma given in Fig. 4(a) are summarized in Table III. Also shown are the exact values of the plasma parameters, the mean value of these parameters for this perturbed sample, and the root mean square (RMS) deviations of the parameters of this sample from the exact values. The RMS deviations of all the geometry parameters are less than 2% , whereas $\bar{\beta}_t$, ℓ_i , and q_1 have the RMS deviations of 5.7% , 9.8% , and 2.2% , respectively.

4.4. Discussion

As was shown in the previous section, the values of $\bar{\beta}_t$ and ℓ_i determined using this approximate method agree to within $\pm 10\%$ and $\pm 15\%$, respectively, with the initial values from the equilibrium calculations and those determined using other methods. There are three possible sources of errors which can contribute to this discrepancy. The first source of errors is associated with the determination of the plasma shape and the poloidal magnetic field along the boundary from the magnetic measurements at some fixed locations surrounding the plasma using the approximate filament plasma current model. The second source of errors arises from the approximate evaluations of the volume-dependent parameters α and R_t . Finally, the third source of errors is associated with the measurement errors of the magnetic signals themselves. The errors arising from the filament plasma current approximation are generally small for most of the plasmas produced in D-III except for those similar to the large-size D-shaped plasma shown in Fig. 4(d), for which it tends to underestimate the elongation E and hence Ω and $\bar{\beta}_t$. These errors are estimated to be less than $\pm 5\%$ of the computed values for the plasma shape parameters and $\bar{\beta}_t$ and $\pm 10\%$ for ℓ_i . The errors that arise from the approximation of α and R_t are estimated to be about $\pm 5\%$ of the computed values for $\bar{\beta}_t$ and about $\pm 10\%$ for ℓ_i . These two sources of errors may be reduced by using a more elaborate physical model to describe the plasma current distribution, which in turn makes possible a more accurate determination of the plasma shapes, α , R_t , and hence $\bar{\beta}_t$ and ℓ_i [8]. For

TABLE III. COMPARISON OF PLASMA PARAMETERS RECONSTRUCTED FROM THE PERTURBED SIGNALS AGAINST THE EXACT VALUES

Case	$a(\text{cm})$	E	$\beta_t(\%)$	$\Omega(\text{m}^3)$	q_t	ℓ_i	$R_\theta(\text{m})$	$Z_\theta(\text{m})$
Exact	37.6	1.35	4.36	5.15	1.52	0.68	1.43	0.81
1	37.3	1.34	4.07	5.02	1.47	0.62	1.43	0.81
2	37.4	1.35	4.40	5.11	1.49	0.60	1.43	0.80
3	37.4	1.37	4.46	5.20	1.52	0.61	1.43	0.81
4	37.7	1.35	4.22	5.18	1.53	0.62	1.43	0.80
5	36.9	1.35	3.78	5.00	1.46	0.78	1.43	0.81
6	37.4	1.33	4.34	5.03	1.47	0.66	1.43	0.81
7	37.3	1.36	4.56	5.14	1.50	0.62	1.43	0.81
8	37.4	1.37	4.46	5.20	1.52	0.61	1.43	0.81
9	37.5	1.34	4.37	5.14	1.51	0.62	1.43	0.81
10	37.4	1.33	4.11	5.06	1.47	0.67	1.43	0.81
Mean	37.4	1.35	4.28	5.11	1.49	0.64	1.43	0.81
RMS	0.30	0.01	0.25	0.09	0.03	0.07	0.00	0.00
RMS %	0.79	1.07	5.66	1.68	2.24	9.83	0.19	0.33

D-III, the measurement errors of the magnetic signals are estimated to be either 3% of the signals or some minimum noise levels. The resulting errors on the computed values of the plasma shape parameters and $\bar{\beta}_t$ are estimated to be about $\pm 5\%$ and to be about $\pm 10\%$ for ℓ_i as discussed above.

A complete calculation for a D-III plasma using this approximate method requires about 3 s of CPU time on the DEC-10 computer. Most of this CPU time is spent in setting up and inverting the 30×30 response matrix $\bar{H}$ and searching for the boundary surface. The response functions G_{kj} are pre-computed and stored in files. However, because the standard deviations σ_k are related to the signal itself and not uniform, $\bar{H}$ cannot be pre-computed and stored. As discussed earlier, the diversity of the plasma shapes attainable in D-III requires the use of an elaborate scheme to obtain the boundary. Both of these factors contribute heavily to the CPU time requirement. Typically, analyses of ten time slices for each shot are performed on line between plasma discharges in D-III.

Appendix A

Evaluation of the volume-dependent parameter α

The parameter α can be conveniently evaluated by transforming the independent co-ordinates to the flux co-ordinates (ρ, θ, φ) , where ρ is a flux surface label varying between 0 and a , θ is a poloidal angle, and φ is a toroidal angle. The transformation is valid providing that the Jacobian $\sqrt{g}$ is finite. B_p and B_z can be expressed in terms of the toroidal current $I(\rho) = I_0 \hat{I}(\rho)$ as

$$B_p = \frac{2I_0}{c} \left(\frac{g_{\theta\theta}}{g} \right)^{1/2} \frac{\hat{I}(\rho)}{\langle g_{\theta\theta}/\sqrt{g} \rangle} \quad (\text{A.1})$$

$$B_z = \frac{2I_0}{c} \frac{Z_\theta}{\sqrt{g}} \frac{\hat{I}(\rho)}{\langle g_{\theta\theta}/\sqrt{g} \rangle} \quad (\text{A.2})$$

where

$$g_{\theta\theta} = R_\theta^2 + Z_\theta^2 \quad (\text{A.3})$$

$$\sqrt{g} = R\tau \quad (\text{A.4})$$

$$\tau = R_\theta Z_\rho - R_\rho Z_\theta \quad (\text{A.5})$$

and

$$\langle \dots \rangle = \int_0^{2\pi} d\theta / 2\pi \dots$$

denotes the poloidal angle averaging operator. Here, the subscripts ρ and θ denote differentiation with respect to these variables. The volume integration of B_p^2 and B_z^2 can then be written as

$$\int_\Omega dV B_p^2 = \frac{16\pi^2 I_0^2 a}{c^2} \int_0^1 dx \frac{\hat{I}^2(x)}{\langle g_{\theta\theta}/\sqrt{g} \rangle} \quad (\text{A.6})$$

$$\int_\Omega dV B_z^2 = \frac{16\pi^2 I_0^2 a}{c^2} \int_0^1 dx \frac{\hat{I}^2(x)}{\langle g_{\theta\theta}/\sqrt{g} \rangle^2} \langle Z_\theta^2/\sqrt{g} \rangle \quad (\text{A.7})$$

where $x = \rho/a$. The flux surface co-ordinates (R, Z) may be represented in Fourier series as [11]

$$R(\rho, \theta) = R_m - \Delta(\rho) - \rho \cos \theta + \sum_{n=2} R_n(\rho) \cos n\theta$$

$$+ \sum_{n=1} \tilde{R}_n(\rho) \sin n\theta \quad (\text{A.8})$$

$$Z(\rho, \theta) = Z_m - \tilde{\Delta}(\rho) + E(\rho)\rho \sin\theta \quad (\text{A.9})$$

Assuming $\Delta(\rho)$, $\tilde{\Delta}(\rho)$, $R_n(\rho)$, $n \geq 2$, $\tilde{R}_n(\rho)$, $n \geq 1$, and $\rho E'(\rho)/E$ to be small $\sim O(\frac{1}{A^2})$, the integrals in Eqs (A.6) and (A.7) can be carried out to give

$$\alpha \cong \frac{2E^2}{E^2 + 1} + O\left(\frac{1}{A^2}\right) \quad (\text{A.10})$$

where $A = R_m/a$.

Appendix B

Evaluation of $\tilde{\beta}_p$ and ℓ_i for the analytic Solovév equilibrium

The flux surface co-ordinates for the Solovév equilibrium (r, z) can be represented parametrically in terms of the poloidal angle θ as

$$r^2(x, \theta) = r_m^2(1 - 2C_1 x \cos\theta) \quad (\text{B.1})$$

$$z(x, \theta) = \tilde{E} x \sin\theta \quad (\text{B.2})$$

where $x = \rho/a$. B_p and B_z can then be expressed as

$$B_p = B_0 x \frac{\beta_{t0} r_m^2 C_1}{\tilde{E}} \left(\frac{C_1^2 r_m^4 \sin^2\theta}{r^2} + \tilde{E}^2 \cos^2\theta \right)^{1/2} \quad (\text{B.3})$$

$$B_z = B_0 x \beta_{t0} r_m^2 C_1 \cos\theta \quad (\text{B.4})$$

The volume integrations of P , B_p^2 , and B_z^2 now yield

$$\int_{\Omega} dV P = \frac{a^3 B_0^2}{4} \frac{\pi}{r_m} \sqrt{\frac{2\beta_{t0}}{\alpha_F}} \quad (\text{B.5})$$

$$\begin{aligned} \frac{1}{8\pi} \int_{\Omega} dV B_p^2 &= \frac{a^3 B_0^2}{4} \pi \tilde{E} \left(\frac{\beta_{t0}}{2} \right)^{1/2} \\ &\times \left\{ 1 + \frac{r_m^2}{\tilde{E}} \left[1 - \frac{1}{6C_1^2} \left(1 - |1 - 4C_1^2|^{3/2} \right) \right] \right\} \end{aligned} \quad (\text{B.6})$$

$$\frac{1}{8\pi} \int_{\Omega} dV B_z^2 = \frac{a^3 B_0^2}{4} \pi \tilde{E} \left(\frac{\beta_{t0}}{2} \right)^{1/2} \quad (\text{B.7})$$

The normalization poloidal magnetic field $B_{pa} = B_{A0}$ is

$$\begin{aligned} B_{pa} &= \frac{1}{\oint_{\Gamma} d\ell} B_0 2\pi a \left(\frac{\beta_{t0} r_m^2 C_1}{2\tilde{E}} \right) \\ &\times \left(\tilde{E}^2 + \frac{2C_1^2 r_m^2}{1 + \sqrt{1 - 4C_1^2}} \right) \end{aligned} \quad (\text{B.8})$$

The plasma volume Ω is found to be

$$\Omega = 2\pi^2 a^3 r_m^2 C_1 \tilde{E} \quad (\text{B.9})$$

$\tilde{\beta}_p$ and ℓ_i are then given by

$$\tilde{\beta}_p = \frac{8\pi \int_{\Omega} dV P}{\Omega B_{pa}^2} \cong \frac{E^2}{E^2 + 1} \quad (\text{B.10})$$

$$\ell_i = \frac{\int_{\Omega} dV B_p^2}{\Omega B_{pa}^2} \cong \frac{1}{2} \quad (\text{B.11})$$

where $E = \tilde{E}/(r_m C_1)$ is the plasma elongation. The parameter α and the line integrals S_1 , S_2 , and S_3 are found to be

$$\alpha \cong \frac{2E^2}{E^2 + 1} \quad (\text{B.12})$$

$$S_1 \cong 2 \left[1 + 0 \left(\frac{1}{r_m^2} \right) \right] \quad (\text{B.13})$$

$$S_2 \cong \frac{3E^2 - 1}{2(E^2 + 1)} \left[1 + 0 \left(\frac{1}{r_m^2} \right) \right] \quad (\text{B.14})$$

$$S_3 \cong \frac{(E^2 + 3)}{2(E^2 + 1)} \left[1 + 0 \left(\frac{1}{r_m^2} \right) \right] \quad (\text{B.15})$$

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